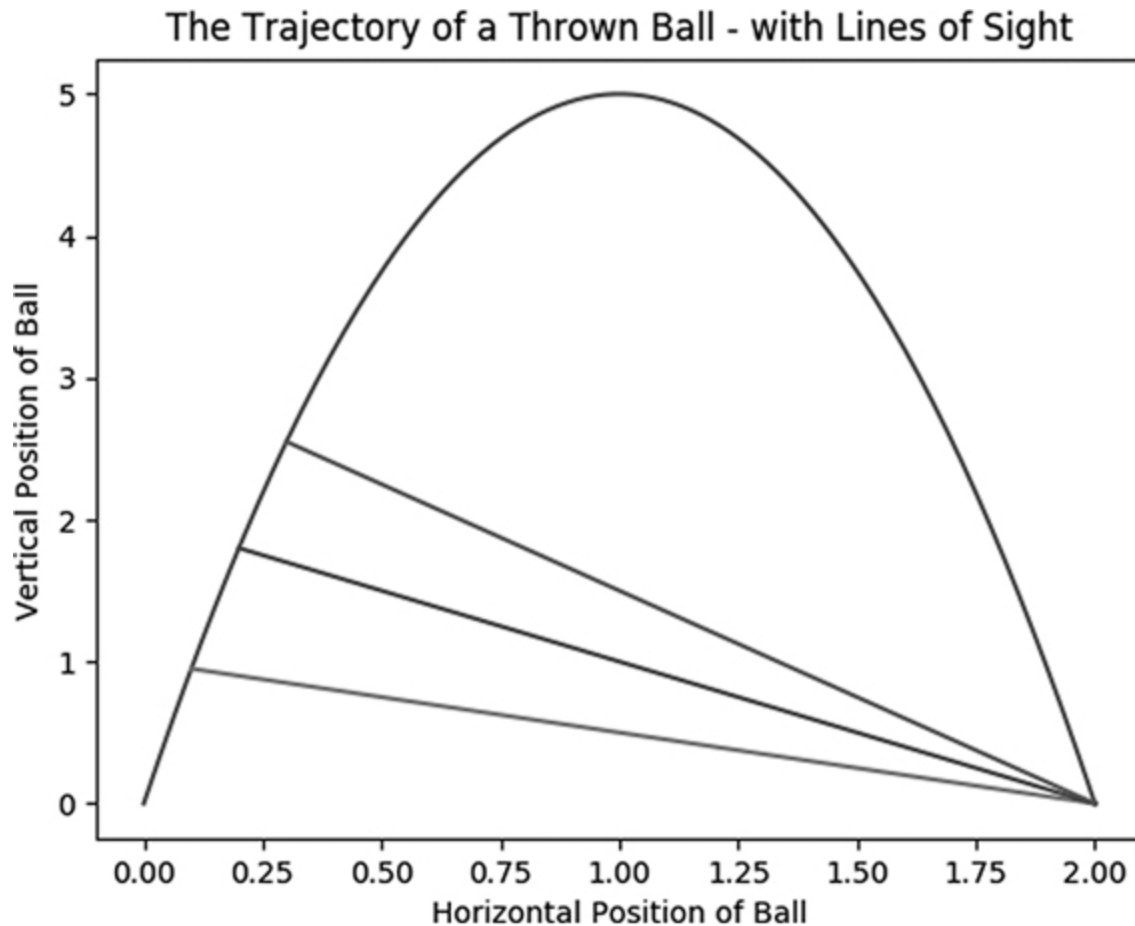




[Figure 1-2:](#) The trajectory of a hypothetical thrown ball, with line segments representing the outfielder looking at the ball as it travels

As the ball progresses through its flight, the angle of the outfielder's line of sight continues to increase, and the outfielder has to keep tipping their head back until they make the catch. Let's call the angle between the ground and the outfielder's line of sight with the ball *theta*. We assume that the outfielder is standing at the ball's eventual destination ($x = 2$). Recall from high school geometry class that the tangent of an angle in a right triangle is the ratio of the length of the side that's opposite the angle and the length of the side that's adjacent to the angle (and is not the hypotenuse). In this case, the tangent of *theta* is the ratio of the height of the ball to its horizontal distance from the outfielder. We can plot the sides whose ratio constitutes the tangent with the following Python code:

```
xs5 = [0.3, 0.3]
ys5 = [0, ball_trajectory(0.3)]
xs6 = [0.3, 2]
```